

Cryogenic Current Steering DAC With Mitigated Variability

M.E.P.V. Zurita, L. Le Guevel, G. Billiot, A. Morel, X. Jehl, A.G.M. Jansen, and G. Pillonnet

Abstract—The paper presents an 8-bit differential current steering digital-to-analog converter (DAC) for the cryogenic front-end of future quantum computers. Recently published characterization of CMOS technology reveals the deterioration of transistor matching properties at cryogenic temperatures. The preliminary study of current mismatch in the FDSOI technology at 4.2 K allows proper sizing of the 255 current-source unit cells to mitigate non-linearities and optimize the static DAC performance. Based on this study we minimize the power consumption while maintaining the targeted non-linearity, i.e. $7.3\mu\text{W}$ for $\text{DNL}=0.64$ LSB. With a 6.6 mV output range and 26 μV voltage step, our DAC is compatible with the foreseen requirements for fine-grain biasing a Si-based qubit matrix at an expected 100MS/s on-chip output drive.

Index Terms—cryoelectronics, quantum computing, digital-to-analog converter, current steering, FDSOI.

I. INTRODUCTION

Quantum computers have the potential to solve problems with unprecedented speed in the field of, e.g., cryptography, data searching, or quantum-mechanical structure analysis [1]. In view of the advantage of a quantum computer with respect to a classical computer [2], it will be necessary to scale-up the number of quantum bits far beyond the tens of qubits reached in existing set-ups.

Apart from the development of quantum-core systems suitable for upscaling the number of qubits, it will be equally important to have the adequate control and read-out electronics for the solid-state qubits operating at sub-1K temperatures [3]–[6]. Using room temperature (RT) electronics involving massive cabling would lead to losses in data-transfer speed, parasitic interference, and thermal load on the low temperature stage. The implementation of integrated electronics at low temperatures offers the opportunity to overcome these problems in the upscaling of a quantum computer.

The realization of Si spin qubit in quantum-dot systems made with industrial FDSOI CMOS technology [7] opened the way to use control electronics made with the same semiconductor technology [8]. Thereby both the quantum core and the control electronics can be placed nearby as shown in Fig. 1 or even fully integrated on-chip at low temperatures. Placing nearby 4.2 K has the advantage of enhanced cooling capacity.

Fig. 1a depicts a Si-qubit system consisting of spin-charge quantum states in a double dot formed under the gates via

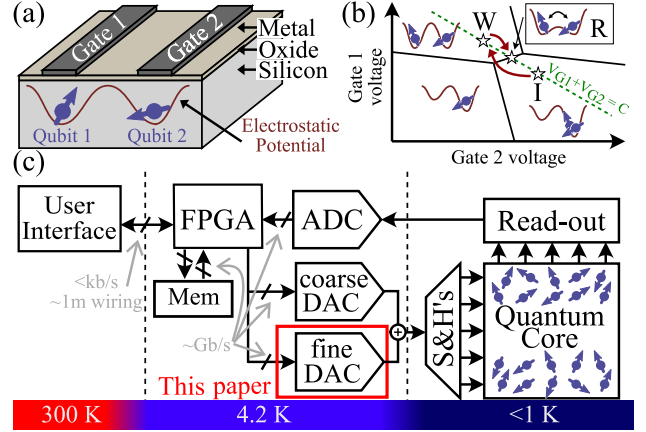


Fig. 1. (a) Quantum dots in MOS structure filled with spin qubit states. (b) Stability diagram of a double quantum dot system with Initialization (I), Writing (W) and Read-out (R) points. (c) Long-term quantum processor interfacing electronics.

applied gate voltages. For typical CMOS structures of nm size at 100 mK, a gate voltage in the mV range is sufficient for incremental electrostatic charging of the dot [8]. The lines of the double-dot stability diagram (Fig. 1b) separate the different occupation states of the two dots with an illustration of a typical set-point sequence of the qubit state.

Thus, a cryogenic DAC of mV range with 8-bit resolution (about $10\mu\text{V}$) allows fine-tuning the qubit state for a typical set-point sequence (Fig. 2b) around a fixed bias-point $\leq 1\text{V}$ [8]. The mW power-consumption of the DACs presented in the literature [9]–[12] limits the scalability in the future quantum core. Recently, [5] realized a low-power DAC showing a high level of non-linearity. To overcome this issue of non-linearity, we present the design of a 8-bit current steering DAC made with 28-nm FDSOI technology operating at 4.2 K with mitigated variability.

II. DAC ARCHITECTURE AND CRYOGENIC DESIGN

The proposed DAC architecture is depicted in Fig. 2. The fully-segmented current steering DAC has been chosen for its inherent capability to operate at high-speed and high-resolution compared to other typical DAC architectures while maintaining reasonable die size and power consumption. Our DAC is composed of 255 unitary current sources I_u driven by 8-bit digital codes received from RT electronics through an on-chip thermometer decoder and a row/column switch array. The non-segmented unitary sources guarantee good dynamic performance, reduced glitches and monotonicity with reasonable die size thanks to the 28-nm node. Each current

Marcos E.P.V. Zurita, Loïck Le Guevel, Gérard Billiot, Adrien Morel and Gaël Pillonnet are with the Université Grenoble Alpes, CEA LETI, Grenoble, France (e-mails: marcos.zurita@cea.fr, loick.leguevel@cea.fr, gerard.billiot@cea.fr, adrien.morel@cea.fr and gael.pillonnet@cea.fr).

Xavier Jehl and Aloysius G.M. Jansen are with the Université Grenoble Alpes, CEA IRIQ/PHELIQS, Grenoble, France (e-mails: xavier.jehl@cea.fr and louis.jansen@cea.fr).

source is driven and summed into two on-chip output resistors R_0 to provide a differential output analog voltage V_{OUTP} . The 1-k Ω polysilicon output resistor offers the targeted 100MS/s bandwidth in on-chip context (<1pF load). We measured 1.1% resistance variation between 300 K and 4.2 K for the polysilicon resistor.

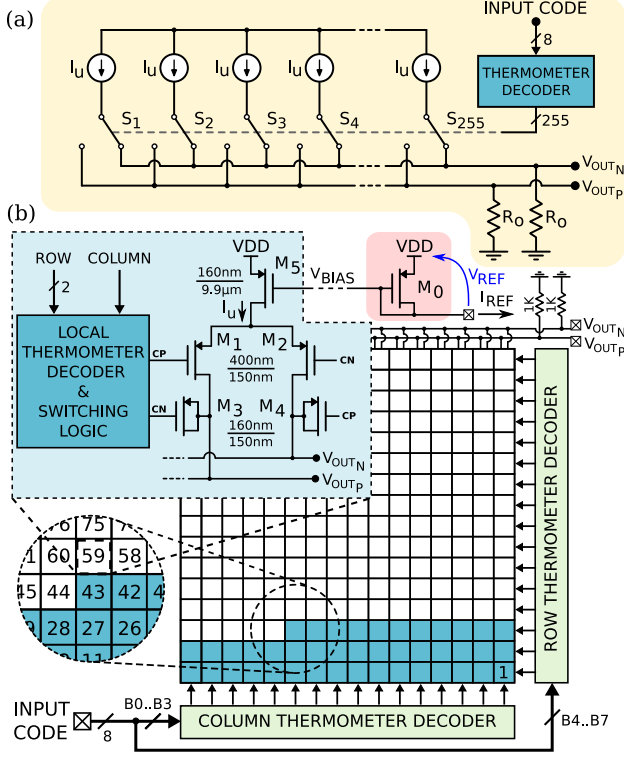


Fig. 2. (a) Operation principle of an 8-bit current steering DAC. (b) DAC architecture with matrix arrangement and transistor-level current-source unit.

The regular 16×16 current source units are arranged in a $168\mu\text{m} \times 213\mu\text{m}$ checkerboard pattern. In each current unit, M_5 provides current from a global reference given by M_0 biased by an RT source I_{REF} . Transistors M_1 and M_2 direct the I_u current to the differential outputs. Dummies M_3 and M_4 ensure the charge injection compensation. C_N and C_P sequence properly the current source switching without flow discontinuities on M_5 which would make it change its operation regime with penalties on conversion rate. Transistor sizes and current bias are chosen based on the analysis conducted in the next section.

By maintaining all current sources polarized at constant DAC power consumption, the thermometer code is essential to provide an effective conversion rate in the order of 100 MS/s for on-chip driving of, e.g., the qubit matrix, and improves robustness against variability-related non-linearities. The digital part, synchronizing the current source switching of the array, is composed of full-custom logic gates made of thin-oxide (GO1) 1V power-supply transistors to minimize the cell area and power dissipation. The analog circuitry uses thick-oxide transistors (GO2) compatible with an extended voltage range of 1.8 V.

III. MISMATCH MITIGATION

The inherent mismatch effects on DAC linearity can be mitigated by proper investigation of the technology and becomes of upper importance at cryogenic temperatures where the mismatch is known to significantly increase [13]. In the absence of cryogenic compact models, we extract the current mismatch from previously acquired experimental data to design the DAC.

For current steering DACs, the main source of non-linearity comes from the current mismatch of the unit-cell current sources (transistor M_5 in Fig. 2b). The DNL variability $\sigma(DNL)$ for a fully-segmented DAC is strictly equal to the total current mismatch $\sigma(I_{ds})/\langle I_{ds} \rangle$ as only one current source state is affected per LSB change. For each gate-source voltage V_g , $\sigma(I_{ds})$ denotes the drain-source-current standard deviation and $\langle I_{ds} \rangle$ is the mean drain-source current.

Once integrated, $\sigma(DNL)$ leads to an estimated maximum variability of INL somewhere mid-code given by $\sigma(INL)_{MAX} = \sigma(I_{ds})/\langle I_{ds} \rangle \times \sqrt{2^N}/2$ with $N = 8$ the DAC resolution bits [14]. $\sigma(I_{ds})/\langle I_{ds} \rangle$ is extracted as a function of V_{gs} at a drain-source voltage V_{ds} of 0.9 V. Statistics are acquired in prior work (partially reported in [8]) within a set of 24 thick-oxide P-type MOSFETs for each of 7 different L/W geometries with lengths L of 0.16, 0.625, and $5\mu\text{m}$ following a similar procedure as in [15]. From 300 to 4.2 K, the total current mismatch can be modeled by the following improved Croon model for modern nodes [16]:

$$\left(\frac{\sigma(I_{ds})}{\langle I_{ds} \rangle} \right)^2 = \left(\frac{g_m}{I_{ds}} \right)^2 \sigma_{V_{th}}^2 + (1 - g_d R_{ds})^2 \left(\frac{\sigma_\beta}{\langle \beta \rangle} \right)^2 + g_d^2 \sigma_{R_{ds}}^2$$

with $g_m = dI_{ds}/dV_{gs}$ the transconductance, β the transconductance gain factor, g_d the total channel conductance, R_{ds} the drain-source access resistance, and σ_X the total mismatch of the quantity X .

The model with free parameters $\sigma_{V_{th}}$, $\sigma_\beta/\langle \beta \rangle$, R_{ds} and $\sigma_{R_{ds}}$ is fitted to the $\sigma(I_{ds})/\langle I_{ds} \rangle$ for a typical long-channel device with $L = 5\mu\text{m}$ as a function of V_{gs} then plotted in Fig. 3 against the normalized current $I_{ds}^\square = I_{ds} \times L/W$. The current source should operate in Moderate Inversion (MI) for $I_{ds}^\square$ between $I_{min}^\square = 0.1$ and $I_{max}^\square = 10\mu\text{A}/\square$ to benefit from a high conductance, to reduce power consumption and to avoid cryogenic variable subthreshold anomalies in weak inversion known to increase the current variability [13]. In MI, mismatch is dominated by the threshold voltage variability whose contribution is shown by the dashed line in Fig. 3. The current mismatch at 4.2 K becomes more sensitive to $\sigma_{V_{th}}$ as g_m/I_{ds} significantly increases from RT.

The Pelgrom-like plots of the cryogenic current mismatch as a function of $1/\sqrt{WL}$ are shown at $I_{min}^\square$ and $I_{max}^\square$ in, respectively, Fig. 4a and b. As expected [16], $\sigma(I_{ds})/\langle I_{ds} \rangle$ follows a linear Pelgrom-like law for small-area devices and becomes constant for large-area devices. Mismatch increases for all geometries by more than $\times 3.5$ from 300 to 4.2 K, highlighting the importance of mismatch mitigation at cryogenic temperatures. Increasing the device area above $1\mu\text{m}^2$ does not result in a significant improvement on current matching [15]. The current source area $L \times W$ is set to $1.5\mu\text{m}^2$ to minimize

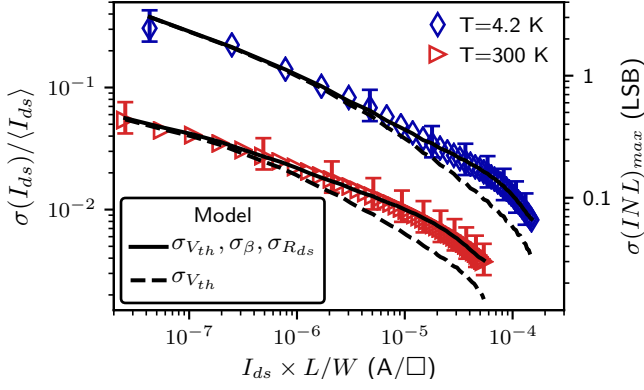


Fig. 3. Normalized current standard-deviation of a PMOS device with $L=5\mu\text{m}$ and $W=1.25\mu\text{m}$. Black lines represent the different contributions from threshold voltage V_{th} , current gain-factor β , and access resistance R_{ds} .

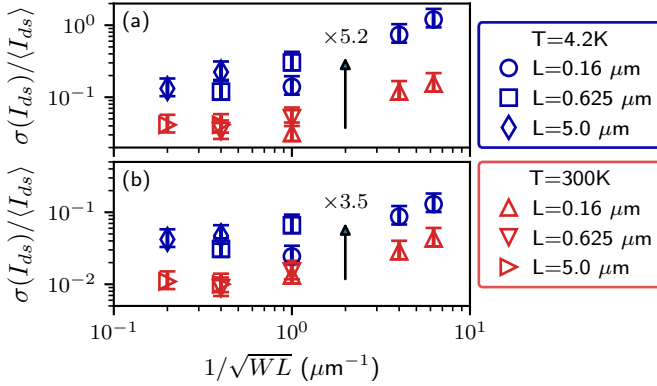


Fig. 4. Normalized current standard-deviation at 4.2 and 300 K of FD-SOI thick oxide PMOS MOSFETs extracted at $I_{ds}^{\square}$ equal to (a) $I_{min}^{\square}$ and (b) $I_{max}^{\square}$. Error bars indicate 90%-confidence interval. Arrows show averaged increase in mismatch from 300 to 4.2 K.

total DAC area while being within the lowest achievable mismatch plateau.

We set the aimed $\sigma(INL)_{MAX}$ at 2 LSB to operate within the aimed low power budget $< 10\mu\text{W}$. This value does not prevent the correct use of the DAC as a 2 LSB voltage is of the order of the characteristic size of the aimed features (star markers in Fig. 1b) and only occurs for a few input codes. The required $I_{ds}^{\square}$ of $1\mu\text{A}$ combined with the desired I_u of 12nA gives a L/W of 83.

Aware of the cryogenic technology mismatch in minimizing the area and non-linearities of the DAC at cryogenic temperatures, we finally choose M5 with a length $L = 9.9\mu\text{m}$ and a width $W = 0.16\mu\text{m}$. The obtained dimensions are sufficiently close to the investigated devices with $L = 5\mu\text{m}$ in our statistical analysis to justify our approach.

According to this design analysis, our sizing has been done to achieve an $\sigma(INL)_{MAX}$ of $0.18 + 0.07/-0.04$ LSB at 300 K and of $1.1 + 0.4/-0.2$ LSB at 4.2 K, so a mismatch increase of 6x, which is experimentally confirmed in the next section.

IV. MEASUREMENT RESULTS

The DAC die occupies $190\mu\text{m} \times 220\mu\text{m}$ including output resistors and thermometer decoders and is mounted on a 48-

pin QFN package (Fig. 5). The measurements are carried out in a liquid Helium dipstick cooled at 4.2K as shown in Fig. 6.

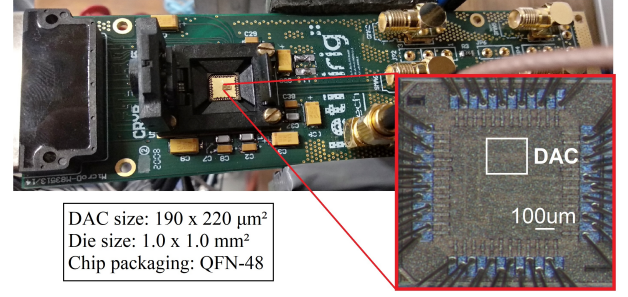


Fig. 5. PCB with DAC chip for cryogenic tests and chip micrograph.

Figure 7 shows the static response at 4.2 K over 6.6 mV differential output range and $26\mu\text{V}$ voltage-step with a total static power consumption including digital part of $7.3\mu\text{W}$. The cryogenic operation degrades the DAC linearities (DNL, INL) as shown in Fig. 8. Upon lowering the temperature down to 4.2 K, the DNL rises by $\times 5.8$ up to 0.64 LSB consistent with the mismatch analysis for the averaged increase in previous section. The INL_{MAX} increases by $\times 4.7$ to 2.96 LSB around mid-code but stays below 2 LSB for most of the input codes. With the DNL measurements being consistent with the mismatch analysis, the relative high INL_{MAX} still smaller than the estimated $3\sigma(INL)_{MAX}$. The increase of variability in DNL at low temperatures is highlighted by the distribution of the 255 sources in the histogram plotted in Figure 9. The reported variabilities $\sigma(DNL)$ at RT and cryogenic temperatures are in line with the given technology characterization in Fig. 4 at 300 and 4.2 K.

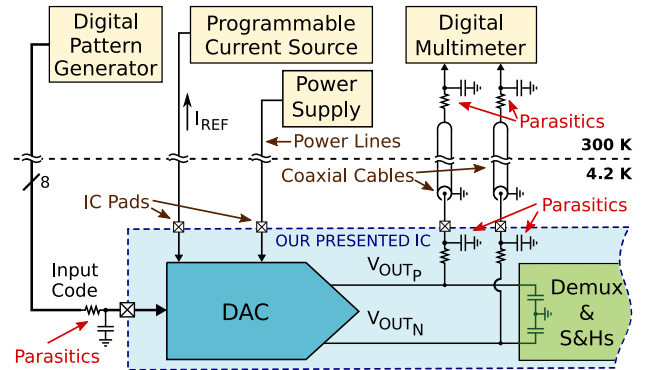


Fig. 6. Cryogenic testbench for DAC characterization.

Measurements at room temperature reveal a LSB-precise full-scale rising time of 350ns with a $1\text{M}\Omega \parallel 35.2\text{pF}$ load indicating a rising time of 9.9ns with 1pF load, which is consistent with our targeted sampling rate of 100MS/s .

V. COMPARISON TO PRIOR-ART AND CONCLUSION

Table I gives a comparison to previously reported cryogenic DACs. Even if mismatch deterioration remains an issue at low temperature, our measured DAC non-linear performances (INL, DNL) are in line with the described design procedure

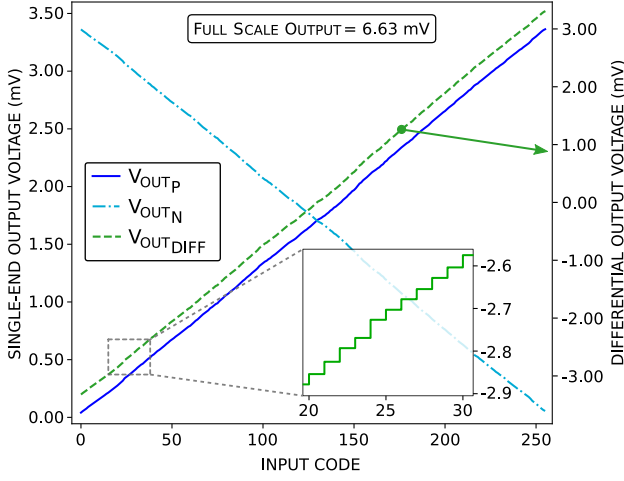


Fig. 7. DAC static response at 4.2 K.

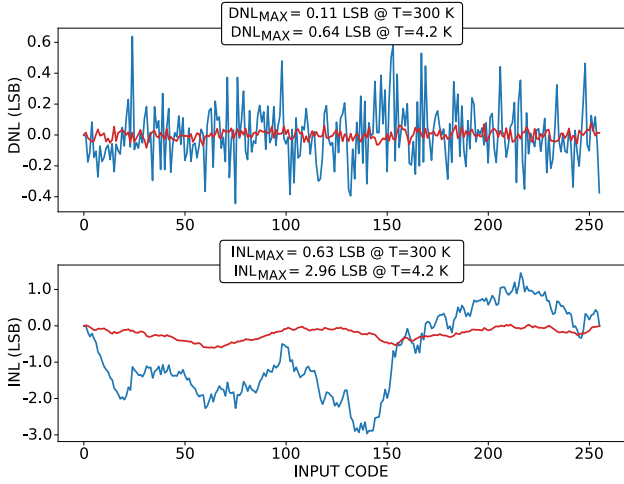


Fig. 8. DNL and INL characteristics at 300 K (red) and at 4.2 K (blue).

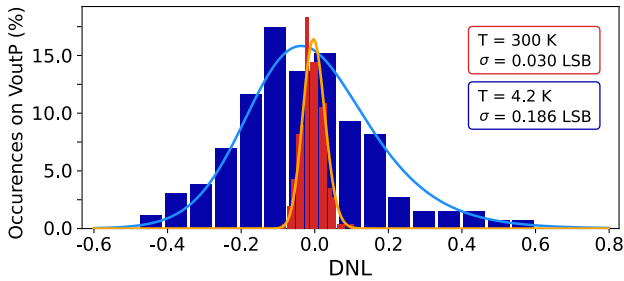


Fig. 9. DNL distribution for the 255 unit current sources at 300 K (red) and at 4.2 K (blue).

given in section III and achieved at 4.2 K without any calibration procedure. The 6.6 mV output range with 8-bit resolution covers the expected requirement to map the gate bias-voltages of a Si-based qubit matrix. The dipstick setup involving cable impedance and input-data communication rate impede us to carry out cryogenic dynamical measurements. Measurements at RT are consistent with a 100 MS/s sampling rate, which is comparable to prior-art. Our DAC achieves the lowest power consumption with an acceptable non-linearity as targeted in the

design stage. The 7.3 μ W static power consumption reported here is compatible with the power budget in the context of a quantum computer operating at cryogenic temperatures and paves the way to low-temperature interleaved and parallel DACs topologies for fast qubit operations.

TABLE I
COMPARISON WITH OTHER CRYOGENIC DACS.

PARAMETER	[9]	[11]	[12]	[10]	This work
Architecture	R-2R	B+H.CS	B+H.CS	H.CS	H.CS
Operating T. (K)	93	4.2	3	4.2	4.2
Technology	0.5 μ m SiGe	0.5 μ m SOS	22nm FinFET	0.5 μ m SOS	28nm FDSOI
Area (mm ²)	0.25	<0.012		<1.1	0.04
Resolution (bits)	8	6	10	10	8
Output Range	200 μ A	6.8mV	≥ 100 mV*	516mV	6.6mV
DNL (LSB)	0.6	0.17		0.35	0.64
INL (LSB)	0.85	0.33	16	4.96	2.96
Conv. Rate (MS/s)	10			100	100**
Consumption (mW)	3.0	<3.2	1.5	32.18	0.007***

* Extracted. ** Estimated. *** Static

B: Binary-weighted; H: Thermometric coded; CS: Current steering.

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